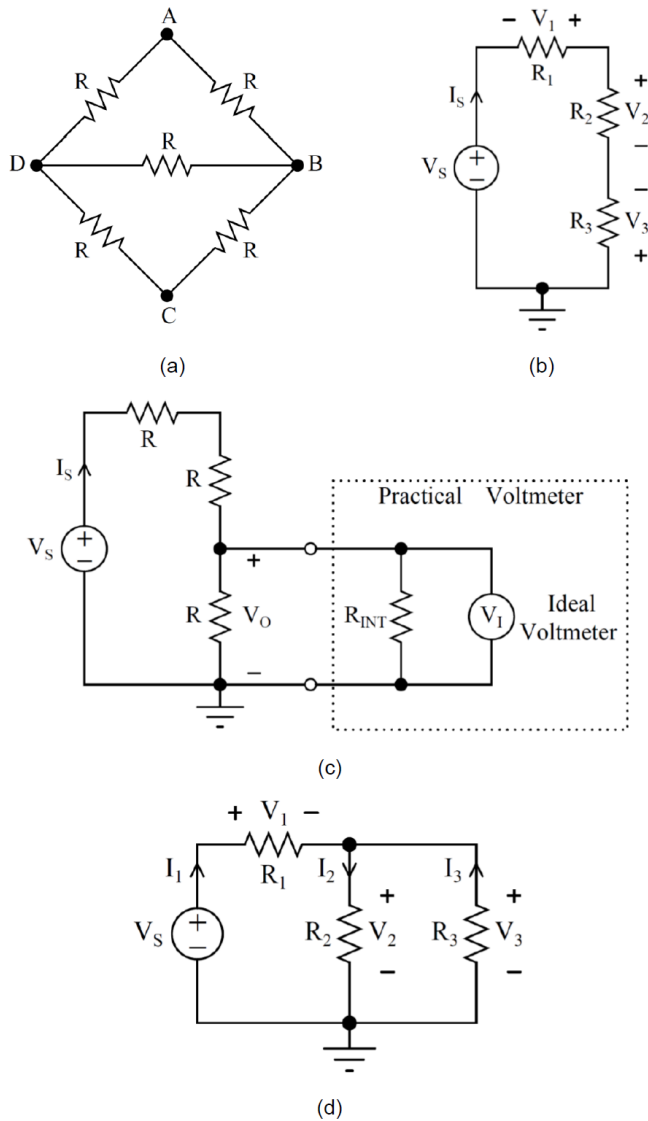


1 Lab 1 Calculations



1.A Equivalent Resistance Calculation

1.A.1 Figure 1-6(a)

1.A.1.a

The equivalent resistance between nodes A & B, R_{AB}

You can find $R_{EQ_{AB}}$ using parallel/series reduction.

$$\begin{aligned}
 R_{EQ_{AB}} &= R_{AB} \parallel [R_{AD} + (R_{DC} + R_{CB}) \parallel R_{DB}] \\
 &= 10k\Omega \parallel [10k\Omega + (10k\Omega + 10k\Omega) \parallel 10k\Omega] \\
 &= 10k\Omega \parallel [10k\Omega + (20k\Omega) \parallel 10k\Omega] \\
 &= 10k\Omega \parallel \left[10k\Omega + \frac{20}{3}k\Omega \right] \\
 &= 10k\Omega \parallel \left[\frac{50}{3}k\Omega \right] \\
 R_{EQ_{AB}} &= \frac{25}{4}k\Omega = 6.25k\Omega
 \end{aligned}$$

1.A.1.b

The equivalent resistance between nodes B & D, R_{BD}

You can find $R_{EQ_{BD}}$ using parallel/series reduction.

$$\begin{aligned}
 R_{EQ_{BD}} &= R_{BD} \parallel (R_{BA} + R_{AD}) \parallel (R_{BC} + R_{CD}) \\
 &= 10k\Omega \parallel (10k\Omega + 10k\Omega) \parallel (10k\Omega + 10k\Omega) \\
 &= 10k\Omega \parallel (20k\Omega) \parallel (20k\Omega) \\
 &= 10k\Omega \parallel 10k\Omega \\
 R_{EQ_{BD}} &= 5k\Omega
 \end{aligned}$$

1.A.1.c

The equivalent resistance between nodes A & C, R_{AC}

$$\begin{aligned}
 R_{EQ_{AC}} &= (R_{AD} + R_{DC}) \parallel (R_{AB} + R_{BC}) \\
 R_{EQ_{AC}} &= (10k\Omega + 10k\Omega) \parallel (10k\Omega + 10k\Omega) \\
 R_{EQ_{AC}} &= (20k\Omega) \parallel (20k\Omega) \\
 R_{EQ_{AC}} &= 10k\Omega
 \end{aligned}$$

1.B Voltage and Current Calculation

1.B.1 Figure 1-6(b)

1.B.1.a

Calculate V_1 , V_2 , V_3 and I_s WITHOUT using the voltage division rule.

Known/Chosen Values

$$\begin{aligned}V_s &= 9v \\ R_1 &= 10k\Omega \\ R_2 &= 20k\Omega \\ R_3 &= 30k\Omega\end{aligned}$$

KVL@ I_s

$$V_1 + V_3 + V_s = V_2$$

Ohm's Law@ R_1

$$V_1 = I_s \times R_1$$

Ohm's Law@ R_2

$$-V_2 = I_s \times R_2$$

Ohm's Law@ R_3

$$V_3 = I_s \times R_3$$

```
% Define symbolic variables
syms is v1 v2 v3

% Define equations
eq1 = v1 + v3 + 9 == v2;
eq2 = v1 == is * 10000;
eq3 = v2 == -is * 20000;
eq4 = v3 == is * 30000;

% Solve equations
sol = solve([eq1, eq2, eq3, eq4], [is, v1, v2, v3])
```

```
sol =
  is: -3/20000
  v1: -3/2
  v2: 3
  v3: -9/2
```

$$i_s = -150\mu A$$

$$v_1 = -1.5v$$

$$v_2 = 3v$$

$$v_3 = -4.5v$$

1.B.1.b

Calculate V_1 , V_2 and V_3 again using the voltage division rule and compare the values with those determined in Part a.

$$V_1 = V_S \times \frac{R_1}{R_1 + R_2 + R_3}$$

$$= V_S \times -\frac{1}{6} = 1.5v$$

$$V_2 = V_S \times \frac{R_2}{R_1 + R_2 + R_3}$$

$$= V_S \times 13 = 3v$$

$$V_3 = V_S \times \frac{R_3}{R_1 + R_2 + R_3}$$

$$= V_S \times -12 = -4.5v$$

1.B.1.c

Is the addition of V_S , V_1 and V_3 equal to the value of V_2 ? Why? Explain your reasoning in detail.

Yes, $V_S + V_1 + V_3 = V_2$. You can verify this by doing the calculation by hand or you can recognize that a KVL@ I_S produces that equation.

1.B.2 Figure 1-6(c)

1.B.2.a

Calculate V_O for $R = 10k\Omega$

$$\frac{9 - V_o}{2R} = \frac{V_o}{R} + \frac{V_o}{1M\Omega}$$

$$\frac{9 - V_o}{20k\Omega} = \frac{V_o}{1k\Omega} + \frac{V_o}{1M\Omega}$$

```
% Define symbolic variables
syms vo

% Define equations
eq1 = (9 - vo)/20000 == (vo/10000) + (vo/10000000);

% Solve equations
sol = solve([eq1], [vo])
```

```
sol =
4500/1501
```

$$V_o = 2.998001v$$

1.B.2.b

Calculate V_O for $R = 10M\Omega$

$$\frac{9 - V_o}{2}R = \frac{V_o}{R} + \frac{V_o}{1M\Omega} \frac{9 - V_o}{20M\Omega} = \frac{V_o}{1M\Omega} + \frac{V_o}{1M\Omega}$$

```
% Define symbolic variables
syms vo

% Define equations
eq1 = (9 - vo)/20000000 == (vo/10000000) + (vo/10000000);

% Solve equations
sol = solve([eq1], [vo])
```

```
sol =
    9/5
```

$$V_o = 4.5v$$

1.B.3 Figure 1-6(d)

1.B.3.a

Calculate I_1 , I_2 , and I_3 WITHOUT using the current division rule.

KCL@ V_2

$$\frac{9 - V_2}{R_1} + \left(-\frac{V_2}{R_3}\right) = \frac{V_2}{R_2}$$
$$\frac{9 - V_2}{10k\Omega} + \left(-\frac{V_2}{30k\Omega}\right) = \frac{V_2}{20k\Omega}$$

```
% Define symbolic variables
syms i1 i2 i3 v2

% Define equations
eq3 = i1 + i3 == i2;
eq4 = i1 == 9-v2 / 10000;
eq5 = i2 == v2 / 20000;
eq6 = i3 == -v2 / 30000;

% Solve equations
sol = solve([eq3, eq4, eq5, eq6], [v2, i1, i2, i3])
```

```
sol =  
v2: 540000/11  
i1: 45/11  
i2: 27/11  
i3: -18/11
```

$$\begin{aligned}v_2 &= 49.\overline{09}V \\ i_1 &= 4.\overline{09}A \\ i_2 &= 2.\overline{45}A \\ i_3 &= -1.\overline{63}A\end{aligned}$$

1.B.3.b

Calculate I_2 and I_3 again using the current division rule and compare the values with those determined in Part a.

$$\begin{aligned}I_S &= \frac{V_S}{R_1 + R_2 + R_3} \\ &= 150\mu A\end{aligned}$$

$$\begin{aligned}I_1 &= I_S \times \frac{R_2 + R_3}{R_1 + R_2 + R_3} \\ I_1 &= 4.1A\end{aligned}$$

$$\begin{aligned}I_2 &= I_S \times \frac{R_1 + R_3}{R_1 + R_2 + R_3} \\ I_2 &= 2.\overline{45}A\end{aligned}$$

$$\begin{aligned}I_3 &= I_S \times \frac{R_1 + R_2}{R_1 + R_2 + R_3} \\ I_3 &= -1.\overline{63}A\end{aligned}$$

1.B.3.c

Is the addition of I_1 and I_3 equal to the value of I_2 ? Why? Explain your reasoning in detail.

Yes, $I_1 + I_3 = I_2$. You can verify this by doing the calculation by hand or realizing that a KCL@ V_2 produces that equation.